

MELROY DCOSTA

Marburg ,Germany |+49 155 108 23806

melroydcosta7@gmail.com

<https://www.linkedin.com/in/melroy-dcosta/>

<https://github.com/melroydco>

M.Sc. Data Science student with hands-on experience in data cleaning, structured data analysis, data validation, SQL, Python, Excel-based reporting, and dashboard development. Skilled in handling large datasets, ensuring data accuracy, and supporting reliable data-driven business processes. Strong interest in master data management, finance operations, data quality, and cross-functional collaboration.

TECHNICAL SKILLS

Programming & Databases: Python, SQL, R, Java, C++

Data Management & Reporting: Master Data Management, Data Quality Checks, Data Validation, Data Cleaning & Transformation, Structured Data Analysis, Business Partner Data, Supplier Data, Product Data, Reporting Workflows

Data Science & Analytics: Exploratory Data Analysis (EDA), Statistical Analysis, Predictive Analytics, Feature Engineering, Data Visualization, Dashboard Development, Data Pipelines, KPI Analysis

Tools & Libraries: Excel, Pivot Tables, VLOOKUP, Power BI, Pandas, NumPy, Matplotlib, Scikit-learn, Jupyter, Git/GitHub, Linux, Docker

Machine Learning & AI: Supervised & Unsupervised Learning, Predictive Modeling, Model Evaluation & Optimization, Deep Learning, Neural Networks, TensorFlow, PyTorch, Keras

AI Systems & Automation: RESTful APIs with Flask, Workflow Automation, End-to-end ML Pipelines, Dialog System Design, Dataset Preprocessing & Validation

WORK EXPERIENCE

AIML Intern, Zephyr Technologies, Mangalore (06/2023-12/2023)

- Cleaned, validated, and structured datasets containing 50,000+ records using Python, Pandas, and SQL-based workflows.
- Performed data quality checks, duplicate handling, missing-value analysis, and transformation of structured datasets.
- Supported reliable data processing workflows by maintaining accuracy, consistency, and completeness of data records.
- Created analytical outputs and reporting-ready datasets to support decision-making and performance tracking.
- Documented data handling processes clearly to improve workflow reliability and team understanding

PROJECTS

RAG-based Knowledge Assistant | Python, LangChain, Vector DBs

- Developing a Retrieval-Augmented Generation (RAG) system integrating vector databases, embedding pipelines, prompt engineering, retrieval evaluation, and agentic workflows for contextual question answering
- Experimenting with document chunking, embedding strategies, prompt engineering, and retrieval evaluation techniques to improve response relevance and system reliability

Gym Management System | PHP, MySQL

- Built a full-stack gym management system to manage member records, trainer details, membership plans, payments, and renewals.
- Designed and maintained a structured MySQL database schema to store customer, membership, billing, and operational data.
- Implemented login authentication, admin dashboard, membership lifecycle tracking, and payment management workflows.
- Ensured accurate record handling by organizing member data, renewal status, payment information, and plan details in a consistent database structure.
- Developed reporting-oriented admin views to support easier monitoring of active members, expired memberships, trainers, and payment records.

Voice-Based Dialog System for Task Automation

- Built end-to-end intelligent dialog system using Python, NLP, and REST APIs for adaptive human-machine interaction
- Integrated ASR, intent detection, and multi-turn dialog management with real-time decision-making logic
- Implemented multimodal interaction pipeline achieving 95% task success rate
- Dockerized system for reproducible deployment and scalability

Restaurant Success Prediction System | Python, XGBoost, Random Forest, Scikit-learn

- Developed a predictive analytics system using Random Forest and XGBoost to analyze structured business data and forecast restaurant success with 85% accuracy.
- Performed data cleaning, data validation, missing-value handling, feature engineering, and exploratory data analysis on large structured datasets.
- Created reporting-ready datasets and dashboard outputs to support KPI analysis, business insights, and data-driven decision-making.
- Applied model evaluation, performance tracking, and SHAP explainability to improve transparency and interpretability of prediction results.
- Organized customer, location, rating, cost, and business performance data into consistent formats for reliable analysis and reporting workflows.

CERTIFICATION

- IBM Data Science Professional Certificate
- Introduction to Supply Chain Finance & Blockchain Technology

EDUCATIONAL BACKGROUND

M.Sc. Data Science

Philipps-Universität Marburg, Germany (10/2024-current)

Relevant coursework: Machine Learning, Big Data Analytics, Database management and NLP, Data analysis, Geodatabases

B.Eng. in Artificial Intelligence and Machine Learning

St Joseph Engineering College, India (07/2020-05/2024)

Languages: English – Fluent ,German – Basic (A2 ,improving)